

Comparison Between 32-Bits Ripple Carry Adder and Kogge-Stone Adder Architectures

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Introduction

Adder architectures importance lies in the impact they have on the operation of multiple circuits used for cell phones, computers, calculators, among many others.

Every mathematical operation elaborated in a binary system will require the elaboration of sums. A better adder architecture means a faster electronic device, that is why researchers constantly seek improvement or development of these circuits. Next, two adder architectures highly used worldwide will be presented in this work.

A Full Adder (Figure 1) consists of a circuit capable of performing the sum of three bits, generally known as A, B and C_{in} .

Multiple Full Adders can be interconnected in cascade to become a Ripple Carry Adder, capable of making sums of a greater number of bits, for example, sums of 32 or 64 bits, among many others.

Although the Ripple Carry Adder architecture presents the great advantage of its easy implementation, its main disadvantage is reflected in the critical path of each Full Adder circuit, representing a problem due to the delay of propagation, since it is required to obtain the last C_{out} to generate the correct sum.

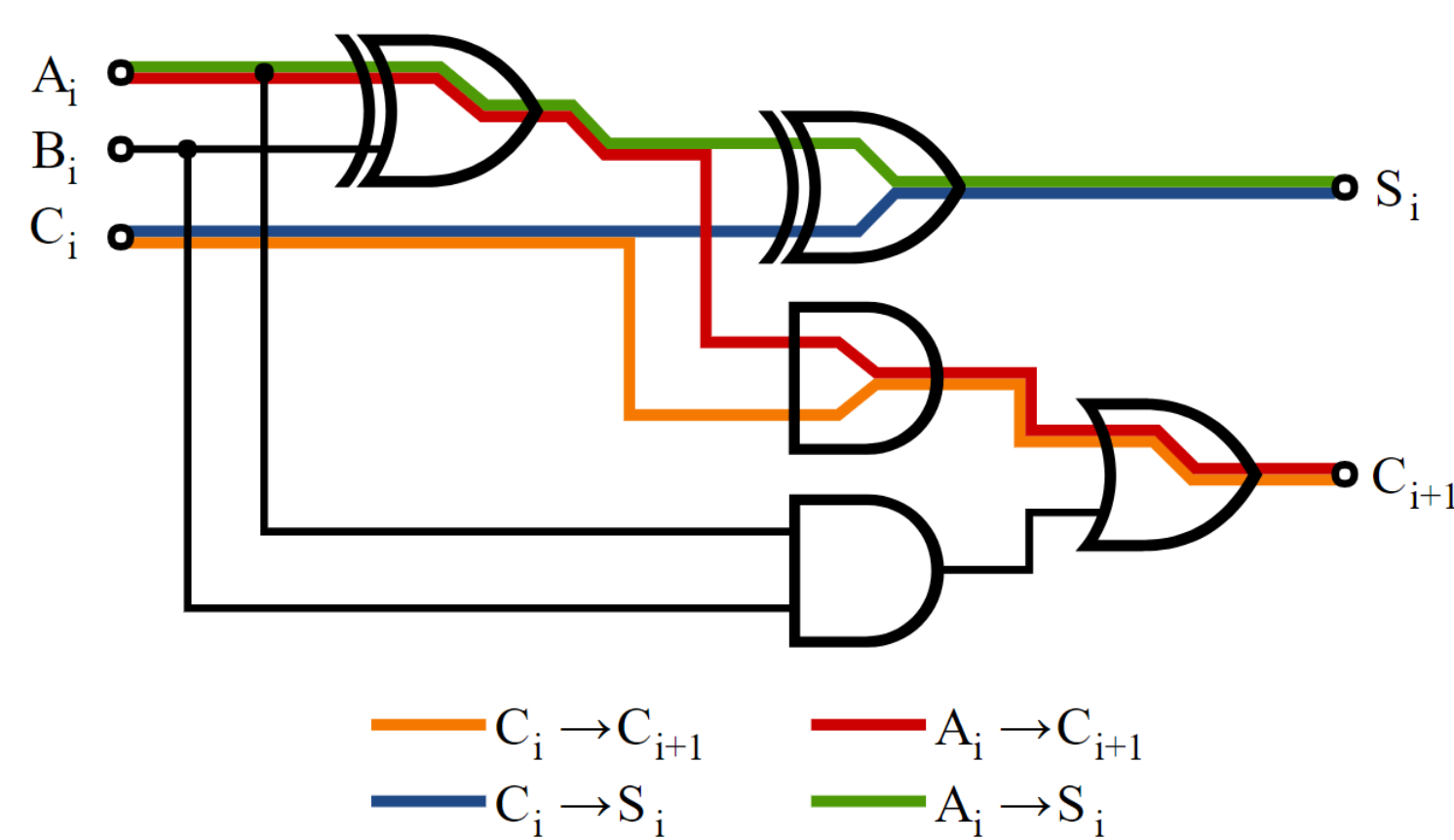


Figure 1: Full Adder Circuit

One solution presented to this problem comes from circuits known as Prefix Adders developed by scientists A. Weinberger and JL Smith [1], the method starts from the idea that even without knowing the value of C_{in} it is possible to assume what will happen at the output of the circuit when performing operations in parallel.

Scientists M. Kogge and Harold S. Stone [2] elaborated the Kogge-Stone Architecture. They use the Prefix Adders principle to elaborate an adder circuit that allows to obtain the final value of the addition in less time.

M. Kogge and H. Stone developed a series of combinations that allow the generation of carry through operations in parallel. The combination of operations developed can be observed in Figure 2.

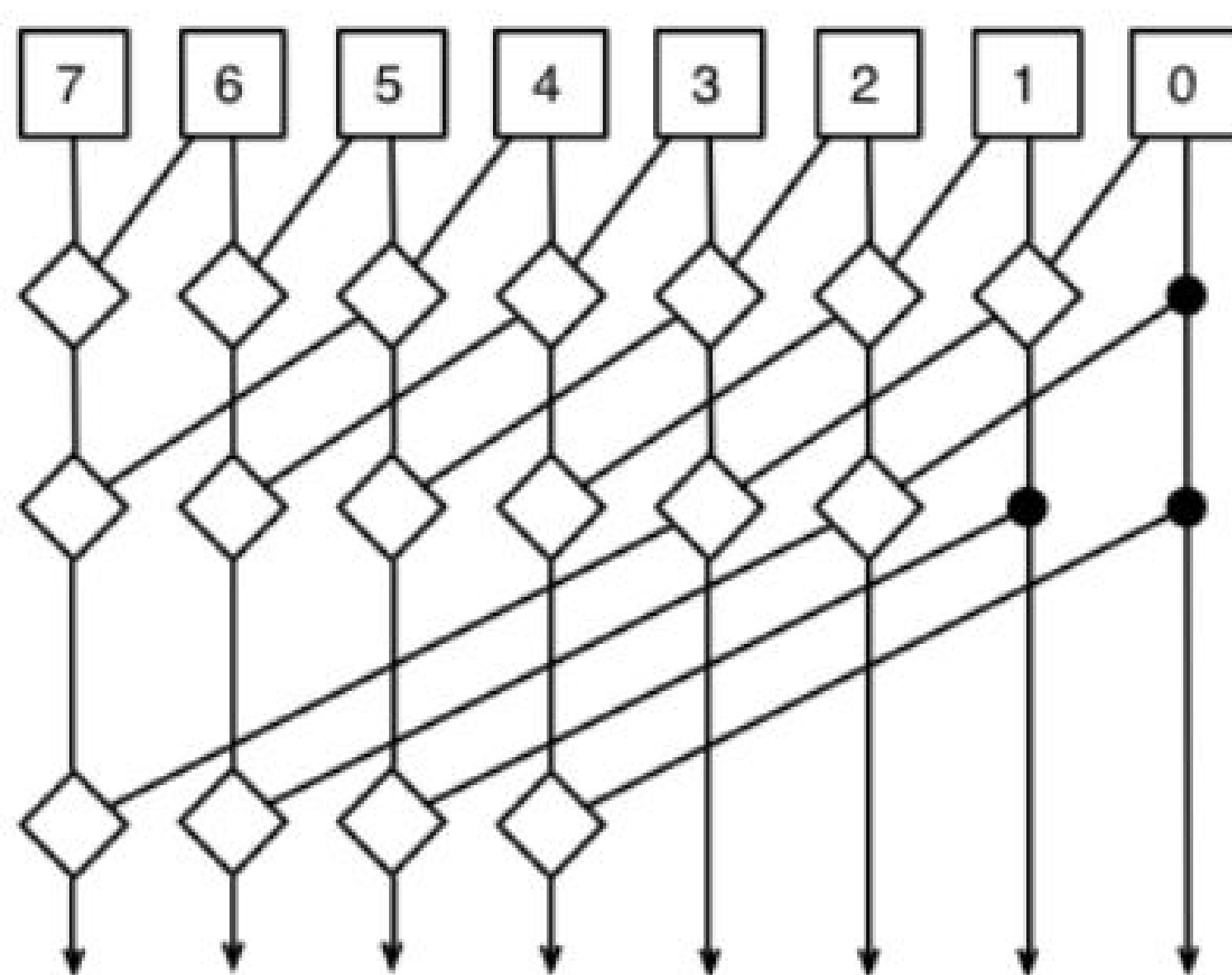


Figure 2: Kogge-Stone combination tree.

Objective

To make a performance comparison between the Ripple Carry Adder and the Kogge-Stone Adder architectures in terms of delay and power consumption.

Methods

Architectures were implemented through modules in the verilog descriptive language. A Flip-flop-based circuit was used to determine the maximum operation frequency, by connecting Flip-flops at the input and output of the architecture under test. Flip-flops were also used for generating clocks, this is represented in figures 3 and 4.

At this point the tool Altium Designer was used to translate the verilog designs for each architecture into mapped hardware designs on FPGA's.

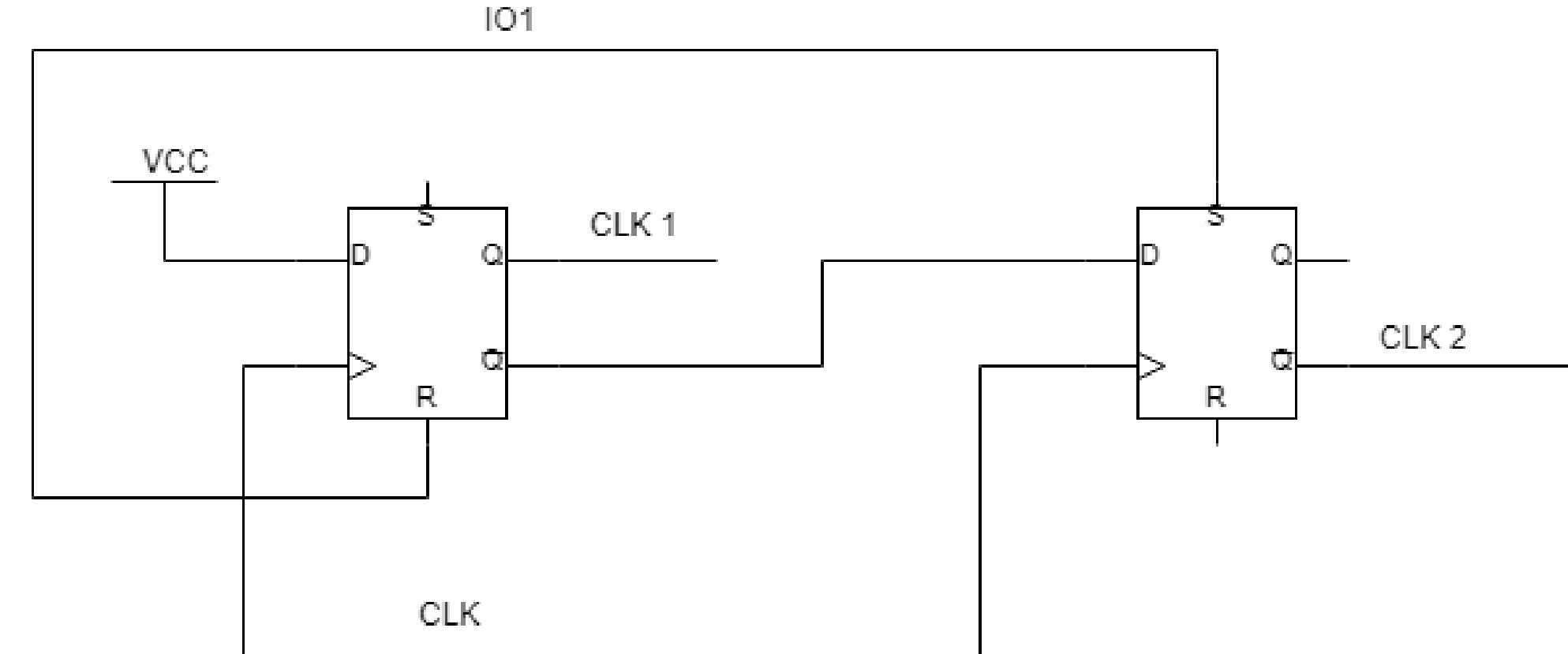


Figure 3: Generation of clock using flip flops

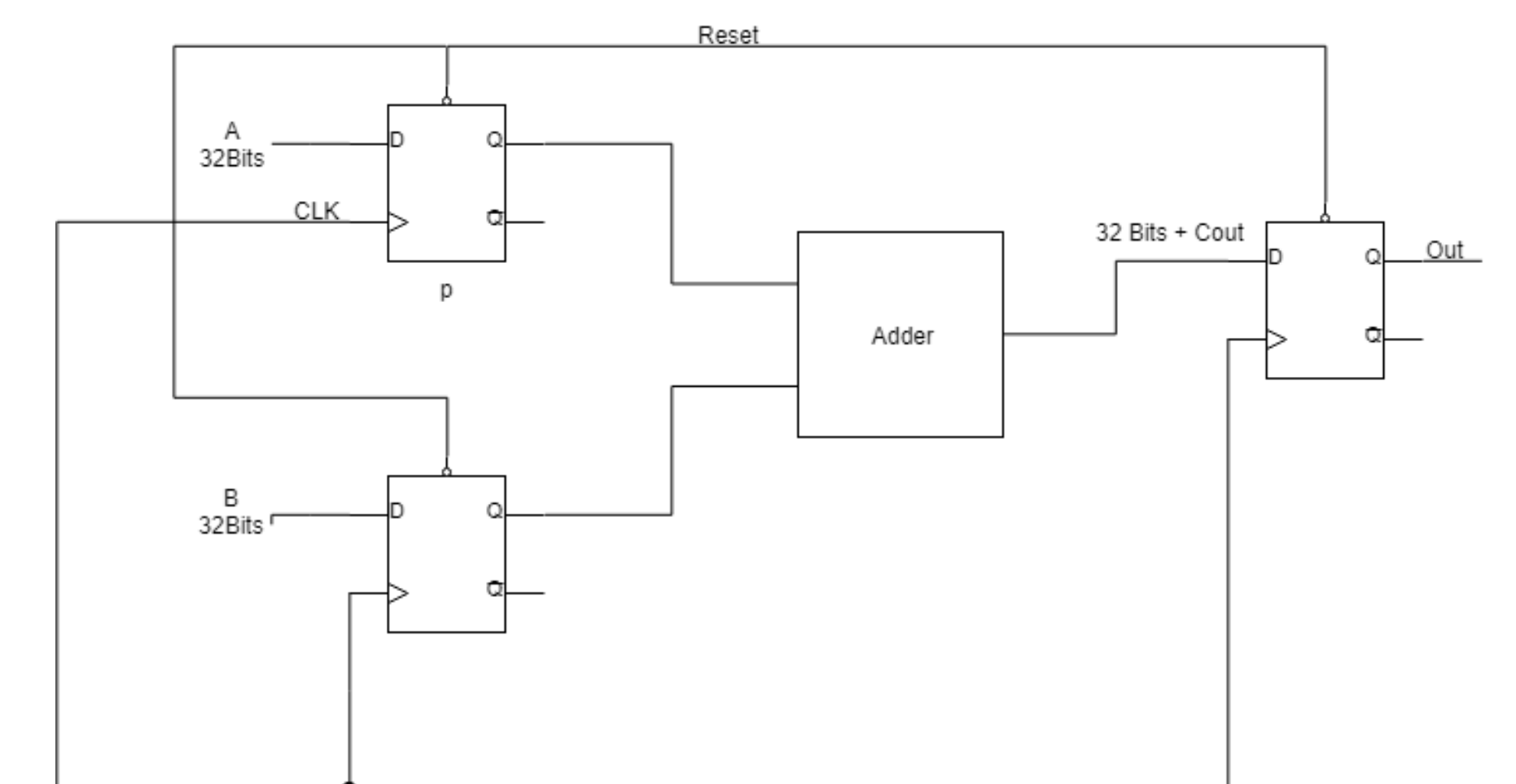


Figure 4: flip flops at the input and output of the architecture adders

Methods

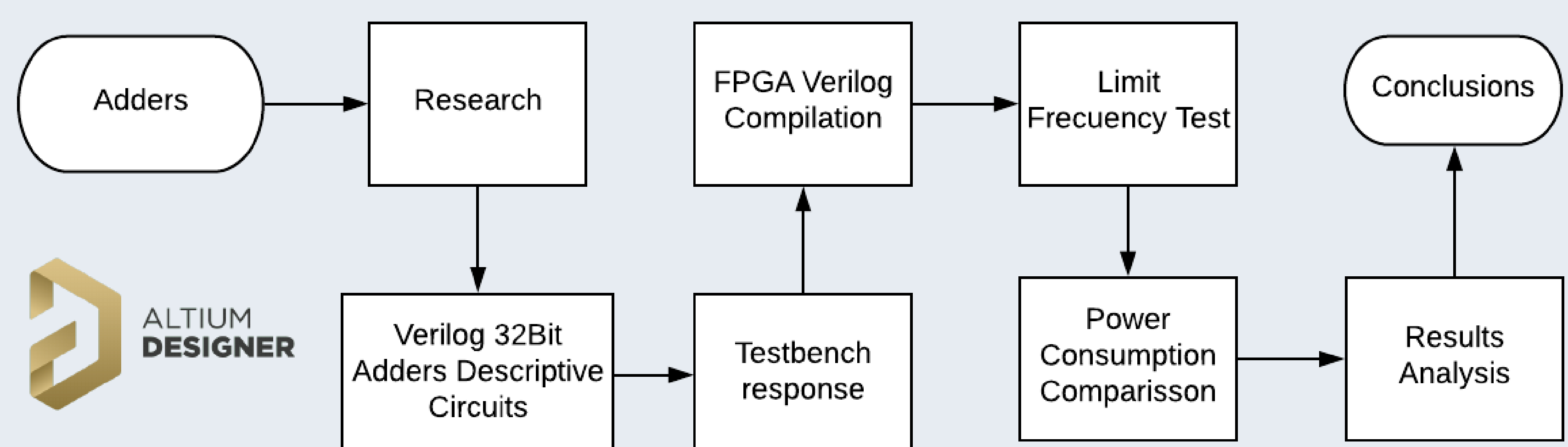


Figure 5: Methodology

Results

By increasing the frequency of operation for the Ripple Carry Adder architecture, a maximum operating frequency of 20 MHz was obtained; Obtaining correct results at this frequency and erroneous results at 25 MHz.

For the Kogge-Stone Adder architecture, increasing the operating frequency resulted in a maximum operating frequency of 100 MHz; Obtaining correct results for this frequency and erroneous results at 150 MHz.

The amount of transistor required for each architecture was determined using the nanoboard mapping output, resulting in 157,440 transistors for the Kogge-Stone Adder, and 125,440 transistors for the Ripple Carry Adder.

In figures 6 and 7 it is possible to observe the power consumption for both the Ripple Carry Adder and the Kogge-Stone Adder respectively.

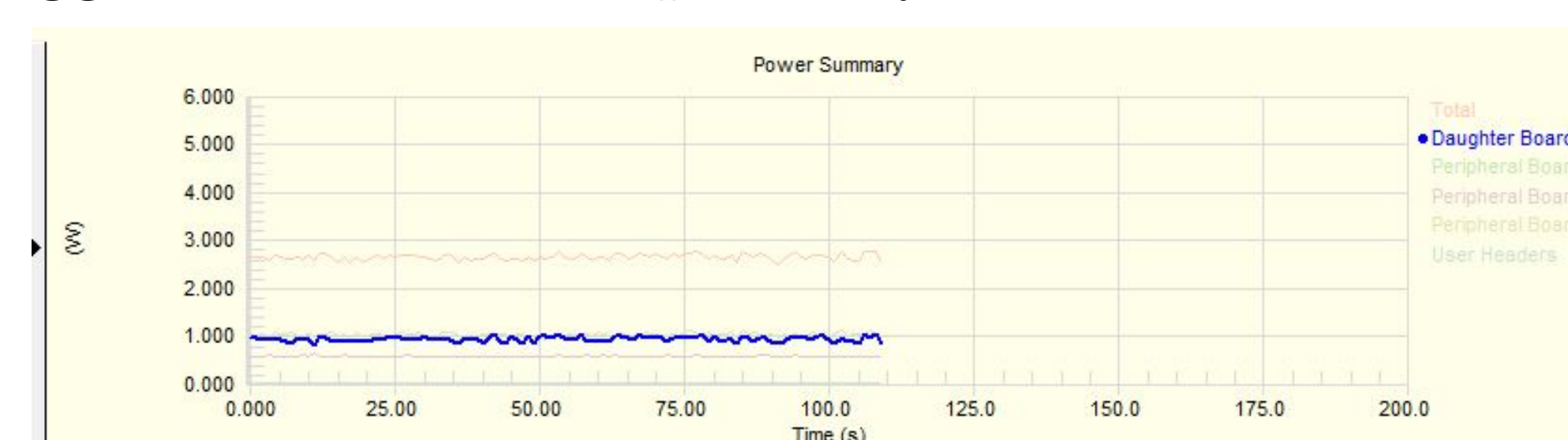


Figure 6: Power in Ripple Carry Adder

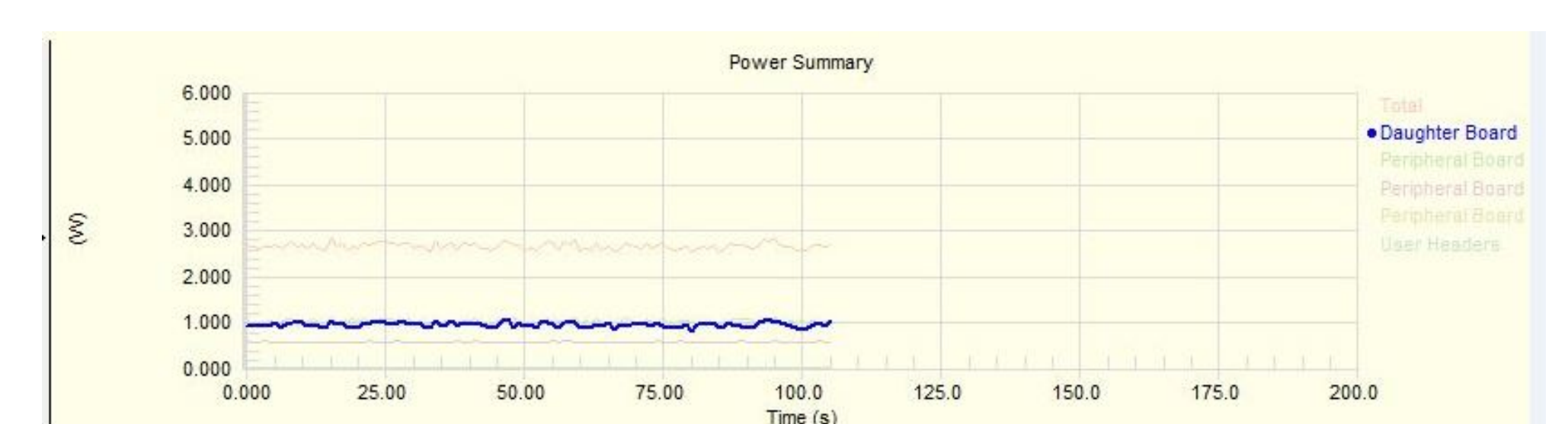


Figure 7: Power in Kogge-Stone Adder

Conclusions and Future Work

- 32-Bit Kogge-Stone Adder architecture showed 5x the performance of the 32-Bit Ripple Carry Adder architecture.
- 32-Bit Kogge-Stone Adder implementation required 25% more hardware compared to the 32-Bit Ripple Carry Adder implementation.
- Power consumption was very similar for both architectures.

In future work we want to determine whether it is possible or not to obtain similar results by implementing the adder circuits presented in multiplier and divider architectures.

References

- [1] A. Weinberger, J. L. Smith. A Logic for High-Speed Addition. 1958.
- [2] Peter M. Kogge, Harold S. Stone. A Parallel Algorithm for the Efficient Solution of a General Class of Recurrence Equations. IEEE Transactions on Computers, Vol. C-22, No. 8. 1973